

A Brief Review on the Effects of the Squat Exercise on Lower-Limb Muscle Hypertrophy

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ABSTRACT

The squat is one of the most widely used exercises in resistance training programs. The aim of this narrative review was to analyze the effect of the squat on lower-limb muscle hypertrophy. Briefly, the available literature indicates that the squat is an effective exercise for inducing hypertrophy of the quadriceps, mainly the vastii, but also the rectus femoris, although to a reduced magnitude. Multiple lines of evidence suggest little to no hamstring hypertrophy from the back squat. Although the gluteus maximus clearly participates mechanically in the back squat, few longitudinal studies exist on the topic. The limited evidence available on this topic suggests deeper squats may be more hypertrophic for the gluteus maximus, and that, squat depth beyond 90° of knee flexion may not provide further hypertrophy of the knee extensors. Despite the popularity of the many squat variations, there are still controversies surrounding their hypertrophic potential for lower-limb musculature. Further studies are needed to investigate the hypertrophic

effects of different squat variations, as well as differences in hypertrophy because of squat depth, stance, barbell position, and different squat apparatuses/machines.

INTRODUCTION

The back squat is a lower-limb multijoint exercise that involves hip, knee, and ankle joints and recruits several lower-limb muscle groups (6,33,34,39). Despite its popularity, the role of this exercise and its variations in promoting hypertrophy remains controversial. For example, regarding back squat depth, a recent study found that squatting with a greater range of motion (ROM) induced greater quadriceps hypertrophy (4), whereas another study did not (21), and both studies suggested that back squat depth affected the level of hypertrophy of certain muscles (4,21). Although some may consider the squat to be enough to induce hypertrophy in all lower-limb muscles, the effect may not be proportional among all muscles involved (4,16,21). Thus, it is important to understand which muscles can be hypertrophied with the squat and its variations and which muscles may

need additional complementary exercises (33,35,42). Moreover, it is possible that not every trainee will perform the back squat properly, and variations may be important to achieve desired benefits (33).

Therefore, this work nonsystematically reviewed the effects of the traditional barbell back squat exercise and its variations on hypertrophy of lower-limb muscles. For this purpose, the relevant literature was retrieved by 3 authors (J.P.N., A.S.R., and E.D.S.) using PubMed and Scopus databases, as well as seminal textbooks, to identify articles that investigated squat training and referred to terms specific to the topic, for example, squat, training, squatting, hypertrophy, muscle size, cross-sectional area (CSA), and muscle thickness (MT). Given that the squat involves the contraction of the primary knee, hip, and ankle extensor muscles, evidence of the effects of the back squat on hypertrophy of these muscles will be considered.

KEY WORDS:

resistance training; exercise selection; muscle growth; inhomogeneous hypertrophy; muscle mass

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SQUAT AND HYPERTROPHY

A summary of the studies included is presented in Table 1. In the following sections, the effects of the squat on measures of muscle hypertrophy on quadriceps femoris, hamstrings, gluteus maximus, and triceps surae are discussed. All studies that analyzed the barbell squat used the back squat with either free-weight or Smith machine variations. In addition, 2 studies analyzed the belt squat in a flywheel machine. Some studies compared squat training with other exercises or intervention types, but only the squat training groups were considered for discussion (e.g., Nakamura et al. (26) compared squat versus squat with isometric rest stretching; Wilson et al. (51) compared squat versus deadlift versus hip thrust).

SQUAT EFFECTS ON QUADRICEPS FEMORIS

Of the 4 heads that make up the quadriceps (vastus medialis, vastus lateralis, vastus intermedius, and rectus femoris), the 3 vastii often present considerably greater muscle excitation than the rectus femoris during the squat (10,11,14,39). This is due to the biarticular nature of the rectus femoris, which, in addition to crossing the knee joint, also crosses the hip joint. However, the rectus femoris does not contribute significantly to knee extension when the hip is flexed, especially at its proximal portion (49). This leads to the notion that the rectus femoris may present a reduced hypertrophic capacity compared with the vastii (12).

In this sense, some studies have analyzed the effect of squat training on quadriceps muscle hypertrophy. Kubo et al. (21), in a sample of untrained young men, observed significant increases in vastii muscle volumes ($+4$ – 7%), assessed by magnetic resonance imaging (MRI), but not in rectus femoris volume ($\sim 0\%$), after 10 weeks of free barbell back squat training performed at either half or full depths ($2\times/\text{week}$, 3 sets of 8–10 repetitions, 60–80% 1 repetition(s) maximum [RM]). Pareja-Blanco et al. (31) examined the effects of 8 weeks of full-depth

free barbell back squat training ($2\times/\text{week}$, 3 sets at ~ 70 – 85% 1RM) at 40 and 20% of the velocity loss threshold in trained young men and observed increases in MRI-assessed volume of the vastus medialis (11 and 12%) and vastus lateralis plus intermedius (9 and 4%), whereas rectus femoris volume remained unaltered (-4% and -1%). Akagi et al. (1) assigned a group of untrained young men to perform 8 weeks of parallel back squat training ($3\times/\text{week}$, 3 sets of 8 repetitions at slow speeds, with 40% 1RM) and observed increases in MRI-assessed volume of the vastii (8–11%), as well as in the rectus femoris, however, of a reduced magnitude (4%). Rosenberger et al. (36) designed a protocol with untrained young men to perform parallel Smith machine barbell back squats for 6 weeks (2 – $3\times/\text{week}$, 3 sets of ~ 8 repetitions, $\sim 80\%$ 1RM) and observed significant increases in MRI-assessed vastii CSA ($\sim 10\%$) but not in rectus femoris CSA (1%). Fonseca et al. (16) reported no statistically significant increases in rectus femoris CSA ($\sim 7\%$; assessed at the medial region, using MRI), whereas there were statistically significant increases in the vastii (~ 7 – 13%), after 12 weeks of parallel Smith machine barbell back squat training in untrained young men ($2\times/\text{week}$, 4–9 sets of 6–10RM). By contrast, Wilson et al. (51) analyzed the MT (using ultrasound [US]) of the rectus femoris midportion (57% of the distal-proximal leg length) and vastus lateralis distal portion (36% of the distal-proximal leg length) of untrained young men before and after 6 weeks of full free barbell back squat ($3\times/\text{week}$, 3–5 sets of 4–8 repetitions, 75–85% 1RM) and observed similar improvements in both the vastus lateralis (11%) and rectus femoris muscles (9%).

In a study exploring region-specific quadriceps hypertrophy along the muscle length (33, 50, and 67% of the proximal-distal thigh length), Earp et al. (10) investigated the effects of performing parallel free barbell back squat at a high-load strength training (3 sets of 3–8 repetitions, 75–90% 1RM) versus a fast-speed jump training (5–7 sets of 5–6 repetitions, 0–30% 1RM) in detrained young

men for 8 weeks ($3\times/\text{week}$). The average effects on US-assessed quadriceps CSA were similar between conditions (~ 16 versus $\sim 15\%$), and no groups reached statistically significant increases on the rectus femoris CSA. An interesting finding was that the different squat modalities elicited distinct inhomogeneous hypertrophy. The jump squat training prompted greater hypertrophy in the vastii muscles at the distal portion (proximal: $\sim 12\%$; medial: $\sim 14\%$; and distal: $\sim 20\%$). Also, nonsignificant changes in the rectus femoris were likely greater in the distal portion than in the proximal and medial ones. Contrarily, for the strength-oriented training group, despite the significant increase in the proximal vastus intermedius greater than the jump training group, the average effects on the vastii were similar between proximal (16%), medial (16%), and distal (16%) portions.

Similarly, Bloomquist et al. (4) also found no difference along the leg length in MRI-assessed “front thigh” muscles CSA (increases ranged from 4 to 7%) after 12 weeks of parallel (deep) free barbell back squats ($3\times/\text{week}$, 3–5 sets of 3–10 repetitions) in detrained young men. However, for the quarter (shallow) squat training group, the authors found a proximal-distal effect on increasing “front thigh” muscles CSA, with changes in the more proximal section ($+4\%$) being greater than in the medial (0%) and distal (-2%) portions (4). Moreover, only the full squat group presented significant pre-to-post increases in leg lean mass assessed via dual-energy x-ray absorptiometry (2.0 versus 1.5%). Also, Zabaleta-Korta et al. (54) investigated trained young men (previous experience > 2 years) and analyzed the changes in proximal (25% of the proximal-distal femur length), medial (50%), and distal (75%) portions of vastus lateralis and rectus femoris CSA (using US) after 5 weeks of parallel Smith machine barbell back squat training ($3\times/\text{week}$, 4 sets of ~ 12 repetitions to failure). The authors observed nonsignificant changes in the rectus femoris (proximal: 3%; medial: 1%; and distal: -2%), whereas only the medial

Table 1
Summary of the studies included regarding the squat training effects on hypertrophy of lower-limb muscles

Study	Sample	Squat	Intervention	Muscles assessed	Main effects
Akagi et al. (1)	Untrained young men	Parallel free barbell back squat	8 wk, 3×/wk, 3 × 8 reps at 40% 1RM	Vastus lateralis, medialis, intermedius, rectus femoris, biceps femoris, semitendinosus, and semimembranosus	Increased MRI-assessed MV of the vastii (8–11%) and rectus femoris (4%) but not the hamstrings (~0%).
Barbalho et al. (2)	Trained young women	Full free barbell back squat	12 wk, 1×/wk, 6 × 4–15RM	Anterior quadriceps and gluteus maximus	Increased US-assessed MT of the anterior quadriceps (12%), and gluteus maximus (9%).
Bloomquist et al. (4)	Detrained young men	Parallel versus quarter free barbell back squat	12 wk, 3×/wk, 3–5 × 3–10RM	Front and back thigh	Increased MRI-assessed front thigh MCSA (4–7%), but not the back thigh MCSA (~0.5%) for the parallel squat group, whereas quarter squat presented reduced results in the front thigh (–2 to 4%) and back thigh (~0%).
Earp et al. (10)	Detrained young men	Parallel free versus jump barbell back squat	8 wk, 3×/wk 3 × 3–8 reps at 75–90% 1RM versus 5–7 × 5–6 reps at 0–30% 1RM	Vastus lateralis, medialis, intermedius, and rectus femoris	Increased US-assessed MCSA of the quadriceps for both groups (~15%), with no significant changes in the rectus femoris. Increases in the vastii were ~16% at proximal, medial, and distal portions for the traditional group, whereas jump training elicited greater distal growth (12; 14; 20%)
Fonseca et al. (16)	Untrained young men	Parallel Smith machine back squat	12 wk, 2×/wk, 4–9 × 6–10RM	Vastus lateralis, medialis, intermedius, and rectus femoris	Increased MRI-assessed MCSA of the vastii (~7–13%), with reduced changes in the rectus femoris (~7%).
Illera-Dominguez et al. (19)	Untrained young adults	Parallel flywheel belt squat	4 wk, 2–3×/wk, 5 × 10 reps	Vastus lateralis, medialis, intermedius, rectus femoris, biceps femoris, semitendinosus, and semimembranosus	Increased MRI-assessed MV of the vastii (9–10%), with reduced changes in the rectus femoris (5%) and hamstrings (2%).
Kubo et al. (21)	Untrained young men	Full versus half free barbell back squat	10 wk, 2×/wk, 3 × 8–10 reps at 60–80% 1RM	Vastus lateralis, medialis, intermedius, rectus femoris, biceps femoris, semitendinosus, semimembranosus, and gluteus maximus	Increased MRI-assessed MV of the vastii (4–7%), with no changes in the rectus femoris (0%) and hamstrings (0%). Greater increases in the gluteus maximus MV for the full-depth condition (7 versus 3%).
Nakamura et al. (26)	Untrained young men	Half flywheel belt squat	5 wk, 2×/wk, 3 × 10 reps	Vastus lateralis, medialis, intermedius, rectus femoris, biceps femoris, semitendinosus, semimembranosus, and gluteus maximus	Increased US-assessed MT of the vastii (~6–15%), rectus femoris (3–9%), but not the hamstrings (1%), and gluteus maximus (–1%).

**Table 1
(continued)**

Pareja-Blanco et al. (31)	Trained young men	Full free barbell back squat	8 wk, 2×/wk, 3 × reps until 40 vs 20% of velocity loss at ~70–85% 1RM	Vastus lateralis, medialis, intermedius, and rectus femoris	Increased MRI-assessed MV of the vastii (~9%) but not the rectus femoris (–3%).
Rosenberger et al. (36)	Untrained young men	Parallel Smith machine back squat	6 wk, 2–3×/wk, 3 × ~8 reps at ~80% 1RM	Vastus lateralis, medialis, intermedius, rectus femoris, and triceps surae	Increased MRI-assessed MCSA of the vastii (~10%) but not the rectus femoris and the triceps surae (1%).
Saeterbakken et al. (37)	Detrained young men	Half free versus Smith versus on wobble-board back squat	6 wk, 2×/wk, 3–4 × 6–10RM	Vastus lateralis	Increased US-assessed MT of the vastus lateralis (~5%) for all conditions.
Usui et al. (46)	Untrained young men	Parallel free barbell back squat	8 wk, 3×/wk, 3 × 10 reps at 50% 1RM with fast versus slow tempos	Vastus lateralis, medialis, intermedius, and rectus femoris	Increased US-assessed MT of the medial and distal vastus intermedius (6–9%), distal rectus femoris (10%), but not the vastus lateralis and medialis at any portion (~2%). The fast-tempo group showed no significant increase in the MT of any muscle/portion.
Weiss et al. (50)	Untrained young adults	Parallel free barbell back squat	7 wk, 3×/wk, 4 × 3–5RM versus 13–15 versus 23–25RM	Anterior and posterior thigh	No pre-training and post-training raw values were provided. The authors indicated that a significant increase in the US-assessed anterior thigh MT was observed but not in the posterior thigh.
Wilson et al. (51)	Untrained young men	Full free barbell back squat	6 wk, 3×/wk, 3–5 × 4–8 reps at 75–85% 1RM	Vastus lateralis and rectus femoris	Increased US-assessed MT of the vastus lateralis (11%) and rectus femoris (9%).
Zabaleta-Korta et al. (54)	Trained young men	Parallel Smith machine back squat	5 wk, 3×/wk, 4 × ~12RM	Vastus lateralis and rectus femoris	Increased US-assessed MCSA of the vastus lateralis (~4%) but not the rectus femoris (1%).

MCSA = muscle cross-sectional area; MRI = magnetic resonance imaging; MT = muscle thickness; MV = muscle volume; RM = repetition(s) maximum; US = ultrasound.

portion of the vastus lateralis presented significant increases (proximal: 4%; medial: 6%; and distal: 1%). On average, the rectus femoris improved by 1% and the vastus lateralis by 4% (54). Oppositely, Usui et al. (46) compared 2 groups of untrained young men that performed parallel free barbell back squats (8 weeks; 3×/week; 3 sets of 10 repetitions at 50% 1RM) at slow versus fast tempos. The authors reported significant increases in MT (using US) for the slow-tempo group in the distal portion of the rectus femoris (10%), and in the vastus intermedius at the medial (6%) and distal (9%) portions, but not in the vastus lateralis and medialis at any portion (ranged from ~ ±2%). The fast-tempo group showed no significant increase in the MT of any muscle or portion (46).

After flywheel training mode, Illera-Domínguez et al. (19) analyzed lower-limb muscle volumes using MRI after 4 weeks (10 sessions) of parallel flywheel belt squat training in untrained young adults (2–3×/week, 5 sets of 10 repetitions). Significant increases in the vastus medialis (+10%), vastus lateralis plus intermedius (+9%), and rectus femoris (+5%) were observed. Moreover, Nakamura et al. (26) found significant improvements in rectus femoris MT using US at the distal portion (+9%) but not at the proximal one (+3%) after 5 weeks of half flywheel belt squat training in untrained young men (2×/week, 3 sets of 10 repetitions). Also, increases of ~6–15% were observed for the vastus lateralis, medialis, and intermedius (26). However, it is important to note that flywheel training prompts greater eccentric action compared with traditional isotonic mode (25); thus, given the different region-specific hypertrophic responses between concentric and eccentric loading (17,18), results from the studies with flywheel belt squat should be interpreted with caution for conventional weight-room use (25). Nonetheless, the results seem to be similar among the abovementioned studies that used different variations of exercise apparatus (flywheel belt, free-

weight barbell, and Smith machine barbell). In the same way, a previous study from Saeterbakken et al. (37) analyzed the effect of 6 weeks of free barbell, versus Smith machine-guided barbell, versus free-weight barbell on wobble boards, half back squat training (2×/week; 3–4 sets of 6–10RM) in de-trained young men. Results revealed similar improvements in vastus lateralis MT (US-assessed) between conditions (free: 4%; Smith: 6%; and wobble boards: 4%). Future studies may consider comparing the adaptations between apparatus on other lower-limb muscles.

Together, studies point to a head-specific hypertrophy for the quadriceps femoris muscle, so that the rectus femoris tends to present reduced gains compared with the vastii. Some of the studies indicated significant increases in the vastii but no effect in the rectus femoris (10,21,31,36,54), others showed reduced (half or less) effects for the rectus femoris in comparison with the vastii (1,16,19), some showed similar gains between the heads, but with rectus femoris improving only at the distal portions (26,46), and 1 indicated increases of similar magnitudes between the rectus femoris and one of the vastii heads (51). These contradictory results regarding the rectus femoris are difficult to reconcile but may be related to the different methodological procedures applied among experiments. A notable difference is that most studies that assessed muscle CSA (a 2-dimensional measure) or volume (a 3-dimensional measure) through MRI found no significant effect, whereas some that analyzed MT (a 1-dimensional measure) through US did. Seemingly, the back squat presents a reduced probability of promoting a significant increase in rectus femoris size, and if it occurs, this may be only in some portions of the muscle. Moreover, from the studies that analyzed portion/region-specific hypertrophy, there is no consistent difference for the increases along the leg length or among the 3 vastii heads. This indicates that vastii demonstrate

similar regional-hypertrophic adaptations across different populations when performing a squat exercise.

SQUAT EFFECTS ON HAMSTRINGS, GLUTEUS MAXIMUS, AND TRICEPS SURAE

The hamstrings consist of the semitendinosus, semimembranosus, and biceps femoris muscles. Although all act as knee flexors, the long head of the biceps femoris, semitendinosus, and semimembranosus also act as hip extensors. Because of the biarticular nature of these muscles, their efficiency in hip extension movements, as seen in the squat, depends on the position of the knee. Thus, when the knees are flexed, these muscles are stretched near the hip but shortened at distal portions, and their capacity to extend the hip is then reduced (39,52). In this way, the hamstring muscles are posited to present null or reduced activation in the squat, especially in relation to the quadriceps (33,35). Only a few studies have tested whether hamstrings can be hypertrophied with squat training, and all have reported almost no effect (1,4,19,21,26,50). Kubo et al. (21) observed no change in CSA of the biceps femoris, semitendinosus, and semimembranosus muscles (~0%) after 10 weeks of full and half free barbell back squats. Akagi et al. (1) found no significant change in the biceps femoris, semitendinosus, and semimembranosus muscle volumes (~0%) after 8 weeks of parallel free barbell back squats. Nakamura et al. (26) also reported no significant change in the hamstring MT (+1%) after 5 weeks of half flywheel belt squat training. Illera-Domínguez et al. (19) observed that, after 4 weeks of parallel flywheel belt squat training, MT changes in biceps femoris short (+3%) and long heads (+3%), semitendinosus (+3%), and semimembranosus (+2%) were small in magnitude, despite the increases of ~10% in the vastii volumes. Bloomquist et al. (4) analyzed the “back thigh” muscles’ CSA along the proximal-distal leg length, without distinction of which specific muscles were measured, and reported an average change of +0.5% (ranging from +3%

to -2%) for the back thigh with parallel and quarter free barbell back squat training. Weiss et al. (50) also reported a similar condition after 7 weeks of parallel free barbell back squat training performed at a variety of intensities in untrained young men and women (3×/week, 4 sets of 3–5, 13–15, or 23–25RM). Although no pre-training and post-training raw values were provided, the authors indicated that no significant time effect was observed in the US-assessed posterior thigh MT, despite the significant increases in the anterior thigh MT.

The gluteus maximus is a voluminous muscle because it is the main hip extensor and can be highly recruited during a squat exercise and its variations (20,34). The gluteus maximus is greatly lengthened in the deeper ROM of the back squat, and because of the relation between hypertrophy in elongated muscles (24,28,29,32,38), it is reasonable to assume that the squat can elicit significant improvements in the size of this muscle. However, a limited number of studies have analyzed gluteus hypertrophy after squat training as opposed to the vast amount of literature that acutely examined gluteus excitation with surface electromyography (sEMG) during several variations of squat and hip exercises (6,20,22,23). It is important to note that considerable evidence suggests that there is no relation between sEMG amplitudes and hypertrophy during highly fatiguing conditions (47). For example, the literature indicates greater muscle excitation in higher loads compared with lower loads (40); however, hypertrophy is similar for both conditions (43). Therefore, analysis of longitudinal studies is necessary to verify hypertrophy. In this sense, Kubo et al. (21) observed that the squat promoted a significant increase in gluteus maximus MT (-1%) after 5 weeks of flywheel half belt squat training. Another study (2) also analyzed gluteus maximus MT after 12 weeks of

full-depth free barbell back squat training (1×/week, 6 sets of 4–15RM) in trained young women and observed significant improvements (9%); however, the veracity of its data has been currently questioned in an expression-of-concern article (48), and results should be analyzed with caution. Therefore, the few data from prospective studies suggest that the back squat can promote significant gluteus maximus hypertrophy, but the results depend on the squat variation. Seemingly, performing the exercise in greater depth seems to be more effective in maximizing gluteus maximus hypertrophy, and this is discussed on the following subsection.

In addition, the squat may have limitations for the growth of the plantarflexor muscle group (45), which is composed of the lateral and medial heads of the gastrocnemius, and the soleus. The force exerted on the ankle joint is substantially lower than that produced by the hip and knee extensor muscles during the squat (5,13,39), thus not placing much tension on the plantarflexor muscles. In addition, the squat does not allow a wide ROM for these muscles (39), which may hamper muscle hypertrophy development (30). Our literature search found no squat-only training study that examined plantarflexor hypertrophy. However, the abovementioned work from Rosenberger et al. (36) examined a group of untrained young men who performed a training program that included parallel Smith machine barbell back squats and Smith machine standing partial calf raises (3 sets of ~12–20 repetitions at ~60–80% 1RM). Despite this issue, the authors did not observe any effect in the MRI-assessed triceps surae CSA (+1%). Because of the lack of studies, the literature regarding the effects of squat on this muscle group is limited for drawing strong conclusions, and further investigations are warranted.

INFLUENCE OF SQUAT DEPTH

The squat can be performed through different depths. The 2 most common categorizations are the full-depth squat, which allows for near-maximum hip and knee ROMs (hamstrings must

come into complete contact with the calves), and the parallel squat, which refers to when the thighs are parallel to the lifting surface (9,15). In a full free barbell back squat, knee flexion ROM is 0°–~140° (considering 0° as full knee extension), whereas a parallel back squat has a knee flexion ROM of 0°–~110°–120°. The quarter and half squats refer to knee flexion ROMs of 0°–30°–60° and 0°–90°, respectively (4,9,15,21). Of note, these knee angles represent the traditional free-weight barbell back squat and may be different when performing squats in machines (e.g., Smith or hack). Thus, quarter and half squats can also be considered as quarter and half ROM compared with the full ROM in the respective squat variation. Nonetheless, a frequent question concerns which ROM should be adopted when squatting to optimize the hypertrophy of lower-limb muscles?

Regarding the quadriceps, studies indicate that the greatest excitation in the squat occurs at angles close to 90°–100° of knee flexion, that is, between the half and parallel ROMs (7,14,44). Although there are several studies verifying muscle activity with sEMG comparing different ROMs (7,14,44), only 2 studies have compared different squat amplitudes on hypertrophy of lower-limb muscles (4,21). Bloomquist et al. (4) analyzed 2 groups of untrained young men who performed “deep” (parallel; 0°–120° of knee flexion ROM) or “shallow” (quarter; 0°–60° of knee flexion ROM) free barbell back squats for 12 weeks. The deep squat improved CSA at all proximal-distal sites (average: 6%), whereas the shallow squat induced significant increases at only 2 of 6 analyzed CSA sites of the front thigh with the proximal site (~3%) greater than medial-distal sites (-2%). Kubo et al. (21) compared full (0°–140° of knee flexion ROM) and half (0°–90° of knee flexion ROM) free barbell back squat training schemes for 10 weeks in young men. Similar results were noticed for quadriceps volume between full and half squat groups, but statistically significant superior gains for full ROM squat were observed in

adductors (6 versus 3%) and gluteus maximus (7 versus 2%) muscle volumes (21). Both studies are in accordance regarding the lack of effect on the hamstrings; however, the 2 studies were different concerning quadriceps hypertrophy. This seems to be related to the squat depths compared because Bloomquist et al. (4) analyzed quarter and parallel squats, and Kubo et al. (21) compared half and full squats. Therefore, considering the 4 tested knee flexion ROM in both studies (quarter, 60°; half, 90°; parallel, 120°; and full, 140°), it seems that squat depth can affect quadriceps size increases, so that squatting in a ROM between half and parallel squat seems necessary to promote satisfactory results in the quadriceps, whereas deeper squatting may not confer an additional benefit.

Regarding the gluteus maximus, only 1 study directly analyzed the influence of squat depth on this muscle. As mentioned earlier, Kubo et al. (21) compared full versus half free barbell back squats and observed greater gains for the full-depth condition (7 versus 3%). In addition, it is interesting to note that Nakamura et al. (26) observed no effect on gluteus maximus MT after half squat, and another study (2) observed significant effects with full-depth squat, giving the notion that squat depth may influence the adaptations. In addition, Yasuda et al. (53) submitted healthy older adults to a 10-week low-load blood-flow-restriction lower-limb training, which included inclined leg-press exercise performed up to 125° of hip flexion (this hip angle is similarly obtained with a barbell parallel back squat) and observed an increase of 6% in MRI-assessed gluteus maximus CSA compared with a non-training control group (−2%). Despite considering the differences in the designs among the 3 studies (e.g., concerning exercise selection), taken together, they vaguely indicate that greater hip ROM seems to be preferred for gluteus hypertrophy (2,21,26,53).

It is also important to mention some issues among studies that verified squat

depth. First, it is difficult to standardize squat depth, and squats can look very different between individuals because of anthropometrics and mobility. Second, it is difficult to standardize how far the pelvis was projected back in the full and partial squat. Third, studies rarely report if participants initiated the movement with knees, hips, or both. In this sense, more studies are needed exploring the different squat variations on gluteus maximus hypertrophy to draw strong conclusions. The effects on the gluteus medius are unknown, and further studies are also required.

PRACTICAL CONSIDERATIONS

Coaches and trainees can feel secure in adding squat exercises to their training programs to induce hypertrophy in the quadriceps and gluteus maximus muscles. Performing the squat at a 90°–120° knee ROM may induce optimal effects on vastii muscles, whereas the rectus femoris tends to present reduced effects (compared with the vastii) in any variation. Moreover, preliminary evidence suggests that squatting at full depth may provide additional increases in gluteus maximus size. Nevertheless, the hamstrings and plantarflexors tend not to increase with squat training. Thus, if the aim is to increase the size of these later muscles (i.e., for a “complete lower-body training”), it is necessary to add exercises that target them (35), such as leg curls (24) and calf raises (27).

Furthermore, squats can be performed in different ways concerning the barbell position (e.g., high-bar, low-bar, or front-bar), stance width (e.g., narrow or sumo squat), apparatus (e.g., sissy, hack, v-squat, belt, or Smith machines), and others (e.g., single-leg squat or Bulgarian squat). Further studies are needed to compare the effects of these variations, given we did not find any longitudinal study concerning variations other than back and flywheel belt squat. Seemingly, individuals can include different squat variations in their training schedules when aiming to train the quadriceps and the gluteus maximus muscles, although little

evidence is available on the magnitude of the hypertrophy on each muscle portion for each variation within the squat pattern. Finally, it is interesting to mention that some recent studies have observed that varying exercise selection throughout training sessions may promote slightly greater hypertrophic responses (8,16) and perceived motivation (3) compared with fixed or non-varying exercise selection conditions. Thus, it is recommended that the squat selection variation be an integrated and cohesive approach designed to target each muscle group using sound rationales based on trainee goals and biomechanical principles (41).

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